

6 A mimic octopus can change shape to look like other organisms.

The photographs show a venomous sea snake, a mimic octopus looking like a venomous sea snake and a mimic octopus looking like a flat fish.



venomous sea snake

(Source: Reinhard Dirscherl / Alamy Stock Photo)



mimic octopus looking like a venomous sea snake

(Source: David Fleetham / Alamy Stock Photo)



mimic octopus looking like a flat fish

(Source: Hal Beral / VWPics / Alamy Stock Photo)

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(a) This octopus is adapted to its niche.

(i) State what is meant by the term **niche**.

(1)

(ii) The mimic octopus has the same banding pattern as the venomous sea snake.

Name this type of adaptation.

(1)

(iii) Suggest why the mimic octopus would change its appearance to look like a flat fish and a venomous sea snake.

(2)

Flat fish

Venomous sea snake



(b) The photograph shows a blue-ringed octopus.



(Source: Oksana Golubeva / Alamy Stock Photo)

This octopus produces a toxin called tetrodotoxin. This toxin affects neurones by blocking sodium channels.

The pain-relieving properties of this toxin were investigated in a drug trial.

Stage II of this drug trial compared the tetrodotoxin with the current pain-relieving drug in a group of cancer patients.

(i) Which of the following describes the purpose of the current pain-relieving drug in stage II of the drug trial?

(1)

- ☐ **A** control
- ☐ **B** digitalis
- ☐ **C** placebo
- ☐ **D** solvent



- (ii) Describe the processes that would have occurred before and after stage II in this tetrodotoxin drug trial.

(4)

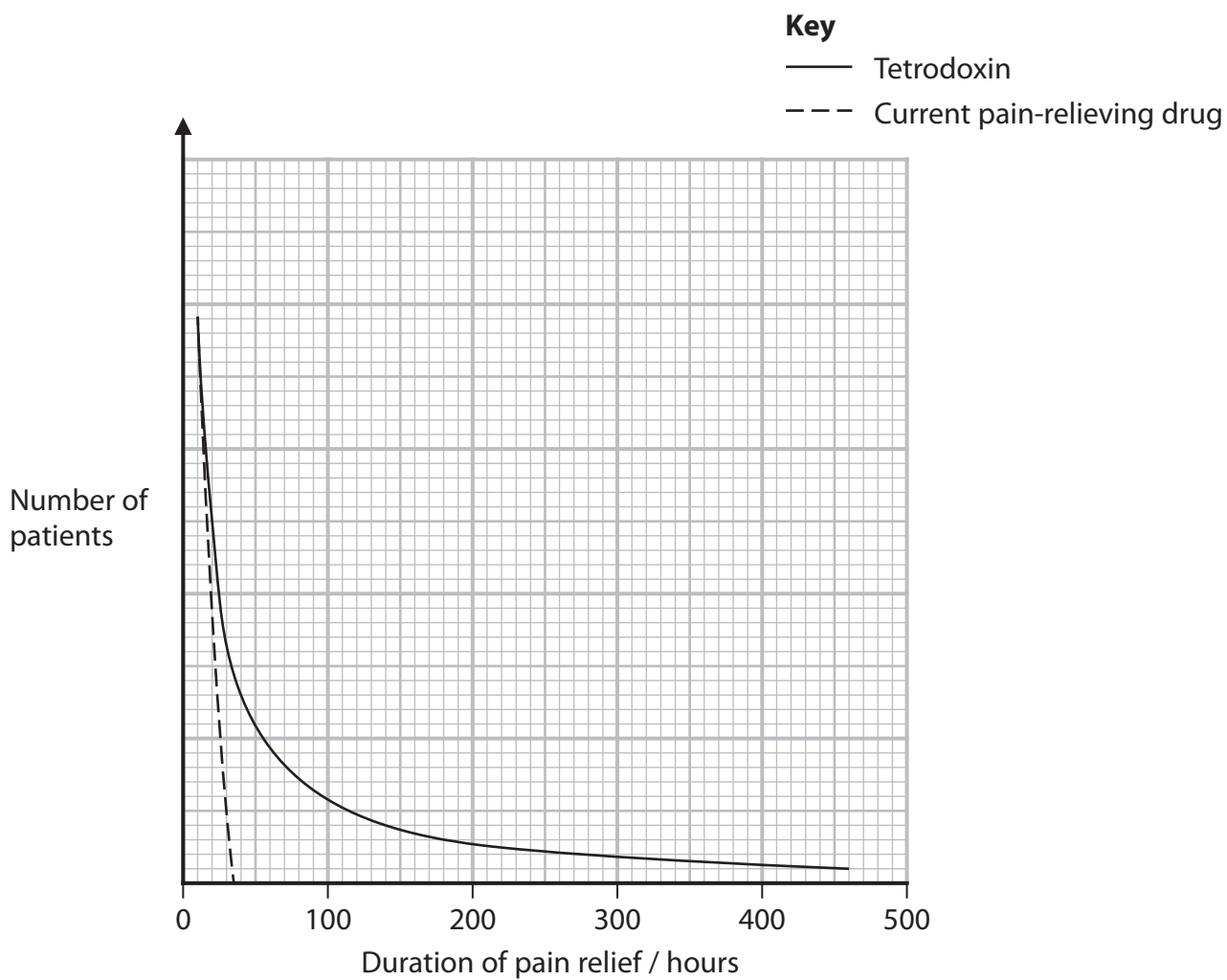
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- (c) The graph shows the duration of pain relief using both tetrodotoxin and the current pain-relieving drug in stage II of this drug trial.



Comment on the results of this investigation.

(4)

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**(Total for Question 6 = 13 marks)**

